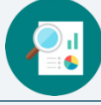
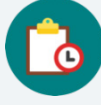


13.5 Exploring Transformations – Horizontal Shifts

Lesson Introduction

 Benchmark	MA.912.F.2.1 Identify the effect on the graph or table of a given function after replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k .		 Learning Targets	- I can discover the effect on the graph of $f(x)$ for a transformation of $f(x + k)$. - I can discover the effect on a table of values of $f(x)$ for a transformation $f(x + k)$.	
 Benchmark Coherence	Building On			Working Toward	
	N/A			N/A	
 Essential Questions	- How can I determine the effect on the graph of $f(x)$ for a transformation of $f(x + k)$? - How can I determine the effect on a table of values of $f(x)$ for a transformation of $f(x + k)$?				
 Lesson Narrative	In this lesson, students explore the horizontal shifts during a transformation of a function. Students begin by exploring shifting the graph of a function. They will then learn about how to write the transformations using function notation.				
 Component Summary	13.5.1 Warm-Up (~5 min.)	Megaship’s estimation activity (<i>SE p. 258</i>)	13.5.3 Guided Instruction (15-20 min.)	Introduction to writing transformations using function notation (<i>SE p. 260</i>)	
	13.5.2 Exploration (10-15 min.)	Exploration activity with shifting the graph of a function (<i>SE p. 258</i>)	13.5.4 Wrap-Up (~5 min.)	Traffic light self-reflection activity (<i>SE p. 262</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms</u> : N/A <u>Additional Terms</u> : N/A		(Optional) Patty paper (Optional) Colored pencils (Optional) Graphing activity		Devices to run Desmos.

 Teacher Tips	Common Misconceptions - Students may interpret the negative sign as a horizontal translation to the left. - Students may interpret the horizontal shift as a vertical shift when writing the function.	Things to Consider 13.5.2 Exploration students are exploring shifting the graph of a function. There are points where students may get confused and struggle. Be cautious not to give away anything during the Exploration, but instead, allow the Exploration to unfold naturally as designed.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect During 13.5.3 Guided Instruction, students are learning about writing transformations using function notation. This is often an area students struggle with initially. Use this instructional time to demonstrate thinking out loud and prompt students to reflect and respond.	MA.K12.MTR.4.1: Discussion and Reflection 13.5.2 Exploration is an activity where students can work with a partner or small group. Part of the Exploration outcomes are for students to explore concepts to make formalized connections with a future activity. During this activity, students will justify results by explaining methods and processes and construct possible arguments based on evidence.
 Differentiate Instruction	Consider the use of digital representations for students to explore as part of this lesson. For example, use Desmos with sliders for students to visualize transformations. Patty paper may also be helpful for students to use when trying to describe movements. Colored pencils are also a nice addition for the students.	
Lesson Notes:		

13.5.1 Warm-Up: Megaships

Component Overview: Students will read about megaships in the Port of Miami and answer two questions.



13.5.1 Warm-Up: Megaships

1. Florida has 15 seaports.

A seaport is where ships can dock to load and unload cargo and passengers. In 2016, the Port of Miami was updated to have a depth of 50 feet so that megaships could dock there. A megaship can hold 406 containers. A normal ship can hold 157 containers.



An article on *JOC.com* said, "Container terminals in gateway ports around the world, already suffering from congestion caused by mega-ships, may not be able to keep up with their ever-increasing size because of limits on crane technology and the lack of yard space."

- a. Based on the capacity of a megaship, why do you think there could be an issue with the lack of yard space?

Answers will vary. There isn't enough space to hold several of them.

- b. A standard shipping container can hold about 24,900 tin cans. How many more tin cans can be delivered by a megaship than by a normal ship?

6,200,100



Leverage this activity as an opportunity for a discussion on the answers and reasonableness. For example, in part a, there isn't an exact answer, but there is enough information available to make a valid argument. Discuss the information available and what can be drawn from that.

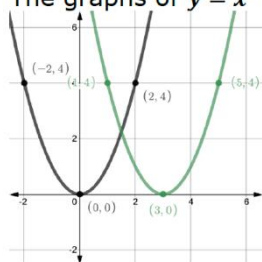
13.5.2 Exploration: Shifting the Graph of a Function

Component Overview: Students will explore the horizontal shifting of the graph of a function. Students will work with a partner to complete two multipart exploration problems.



13.5.2 Exploration: Shifting the Graph of a Function

1. The graphs of $y = x^2$ and $y = (x - 3)^2$ are shown.



- What do you notice about the graphs of $y = x^2$ and $y = (x - 3)^2$?
The vertex of the green graph moved three units to the right of the black graph.
- Work with your partner to describe how to shift the graph of $y = x^2$ so that it coincides with the graph of $y = (x - 3)^2$.
Graph of $y = (x - 3)^2$ shifts three units to the right of $y = x^2$.
- The table of values for $y = x^2$ and $y = (x - 3)^2$ are shown.

$y = x^2$		$y = (x - 3)^2$	
x	y	x	y
-2	4	1	4
0	0	3	0
2	4	5	4

What do you notice when comparing the table of values?

The vertex and the x-values are different in the second table than the first.



Consider using the Desmos link found in the digital area for a quadratic so that students can play with the slider to see the movement for $f(x + k)$.



When circulating the room during this activity, listen for what students are noticing and wondering. While this routine creates a safe entry point for students, it's important that the things that they notice and wonder are focused on the things on the graph that are needed for this Exploration.



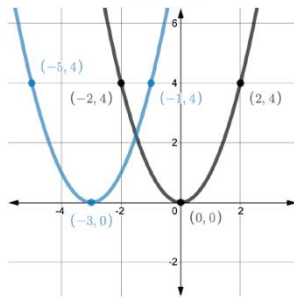
When circulating the room during this activity, listen for what students are noticing and wondering. While this routine creates a safe entry point for students, it's important that the things that they notice and wonder are focused on the values in the table related to the graph work they have been doing in the prior parts.

13.5.2 Exploration: Shifting the Graph of a Function (continued)

- d. Discuss with your partner connections between the x -values in the table of values and the description of shifting the graph of $y = x^2$ so that it coincides with the graph of $y = (x - 3)^2$. Summarize your discussion.

The x -values increased by 3. The green graph shifted 3 units to the right of the black graph (parent function).

2. The graphs of $y = x^2$ and $y = (x + 3)^2$ are shown.



- a. Work with your partner to describe how to shift the graph of $y = x^2$ so that it coincides with the graph of $y = (x + 3)^2$.
The graph of $y = (x + 3)^2$ shifts three units to the left of $y = x^2$.
- b. The table of values for $y = x^2$ and $y = (x + 3)^2$ is shown.

$y = x^2$		$y = (x + 3)^2$	
x	y	x	y
-2	4	-5	4
0	0	-3	0
2	4	-1	4



Listen for the language students are using in part a of this question. If students are making statements such as “It moves to the left,” advance their thinking by asking, “How many units does it shift to the left?” and “How is that represented in the equation or table of values?”

13.5.2 Exploration: Shifting the Graph of a Function (continued)

Discuss with your partner connections between the x -values in the table of values and the description of shifting the graph of $y = x^2$ so that it coincides with the graph of $y = (x + 3)^2$. Summarize your discussion.

The x -values decreased by three. The blue graph shifted three units to the left of the black graph (parent function).



Refer students back to the Desmos graphs. The graphs have sliders where students can manipulate them and see the shifting of the graph happen in response to their values.

c. Complete the statements.

The graph of $y = (x + 3)^2$ is a

- ☐ horizontal shift
- ☐ vertical shift

of the graph of $y = x^2$.

For the graph of $y = x^2$ to coincide with the graph of

$y = (x + 3)^2$, it is shifted

- ☐ down
- ☐ left
- ☐ right
- ☐ up

3 units.

The shift is seen in the table of values by

- ☐ adding
- ☐ subtracting

to each

- ☐ x -value
- ☐ y -value

for each corresponding

- ☐ x -value.
- ☐ y -value.

13.5.3 Guided Instruction: Writing Transformations Using Function Notation

Component Overview: Students will learn how to write transformations using function notation over the course of two multipart problems.



13.5.3 Guided Instruction: Writing Transformations Using Function Notation

The shifts of the graphs can also be written in function notation. The original graph $y = x^2$ can be written as $f(x) = x^2$.

- Notice that the horizontal shift to the right 3 is written as $y = (x - 3)^2$, where the $(x - 3)$ is in place of the x . To write this shift using function notation, substitute $x - 3$ in place of x .

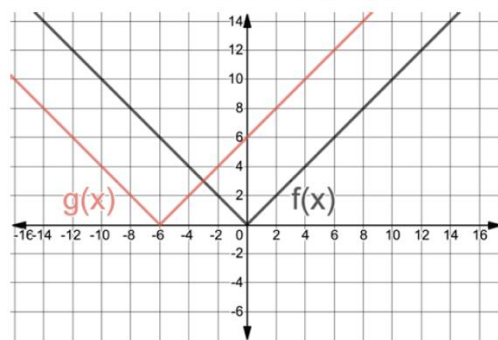
$$f(x) = x^2$$

$$f(x - 3) = (x - 3)^2$$

- Work with your partner to complete the table. The first row has been completed.

Original Equation	Original Function	Shift	Transformed Function
$y = x^3$	$f(x) = x^3$	Right 5	$f(x - 5) = (x - 5)^3$
$y = x^2$	$f(x) = x^2$	Left 7	$f(x + 7) = (x + 7)^2$
$y = \sqrt{x}$	$f(x) = \sqrt{x}$	Right 2	$f(x - 2) = (\sqrt{x - 2})$
$y = 3^x$	$f(x) = 3^x$	Left 1	$f(x + 1) = (3^{x+1})$

- The graphs of $y = f(x)$ and $y = g(x)$ are shown.



For this activity, consider reviewing everything prior to part a as an “I do” modeling opportunity to connect the example with the expression. Then release question 1a for students to work with a partner to complete the table. Review the answers and rationale together as a whole group prior to moving on to question 1b of this activity.



What do you notice about the signs when a function is shifted right vs left?



Use the Desmos link in the digital area for absolute value so that students can play with the slider to see the movement for $f(x + k)$.

13.5.3 Guided Instruction: Writing Transformations Using Function Notation (continued)

How does the graph of $f(x)$ compare to the graph of $g(x)$?

$g(x)$ is shifted 6 places to the left of $f(x)$.

- c. Two function descriptions are shown. Discuss with your partner which function description matches the transformation on the graph. Then, justify your choice.
- $g(x) = f(x + 6)$
 - $g(x) = f(x - 6)$

Justification

The vertex of $f(x)$ moved six units to the left resulting in $g(x)$.

2. The table of values for the function $y = B(x)$ is shown.

x	B
-4	15
-1	8
5	-6
14	-27

- a. Describe the effect that the transformation of $B(x - 2)$ will have on the x -values for each corresponding B -value in the table.

The x -values will increase by 2.

- b. Describe the effect that the transformation of $B(x + 7)$ will have on the x -values for each corresponding B -value in the table.

The x -values will decrease by 7.



Allow students to share out their answers to parts a and b. Prompt students to determine whether they agree or disagree with the answers being shared out. Also determine if there are additional methods or approaches that other students have used.



If the class has time, the teacher can have students experiment with the $y = x$ Desmos link to play with the slider to see the movement for $f(x + k)$.

13.5.4 Wrap-Up: Reflection

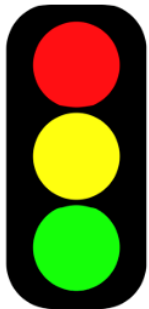
Component Overview: Students will complete a traffic-light reflection activity. They will identify what they are confused about, what they need clarification on, and what they understand.



13.5.4 Wrap-Up: Reflection

1. Describe your learning today by writing a short statement for each.

Stop, I'm confused about ...



Proceed with caution,
I need clarification on ...

Go ahead. I understand ...

Answers will vary.



If students struggle writing a short response, have students mark parts of the lesson as red (confused), yellow (need clarification), or green (understand).